



The Hidden Cost of Higher Education

The Bottom Percentiles

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Motivation

Why this matters:

- College: biggest financial decision for most of students
- Tuition varies significantly, but outcomes are uncertain
- Rankings $\neq$ ROI

Key question: Which colleges actually deliver value relative to cost?





Data Source

<https://collegescorecard.ed.gov/>

:Contains 6,000+ schools analyzed

- Key variables: Net price, completion rate, financial aid, earnings after graduation, etc.



U.S. DEPARTMENT OF EDUCATION
College Scorecard



Data Cleaning & Preparation

1.Column Name Standardization

-Renamed variables using schema (YAML)

```
admissions.sat_scores.75th_percentile.writing:  
  source: SATWR75  
  type: float  
  description: 75th percentile of SAT scores at the institution (writing)  
  index: float
```

2.Feature Selection

-Focused on cost, earnings, aid, and student



Data Cleaning & Preparation

3. Missing data handling

- Dropped highly incomplete columns (>80% missing)
- Filtered schools missing key variables (cost, admissions)
- Imputed missing 4-year earnings using 2022 data

4. Feature Engineering

- Created net_price (public + private average cost)



ROI (Return on Metric)

ROI = Earnings (4 years) / Net price

school.name	school.state	pell_%	earnings	net_price	roi
CUNY Bernard M Baruch College	NY	57.0	82748.0	3033.0	19.7
CUNY Hunter College	NY	56.0	73236.0	2984.0	14.4
CUNY Brooklyn College	NY	57.0	62833.0	3103.0	11.2
CUNY John Jay College of Criminal Justice	NY	60.0	60020.0	3203.0	10.4
CUNY Lehman College	NY	60.0	63353.0	3148.0	10.1
CUNY City College	NY	61.0	67740.0	3776.0	10.0
Trinity International University-Illinois	IL	89.0	51985.0	2835.0	8.4
California State University-Los Angeles	CA	66.0	59756.0	3967.0	8.0

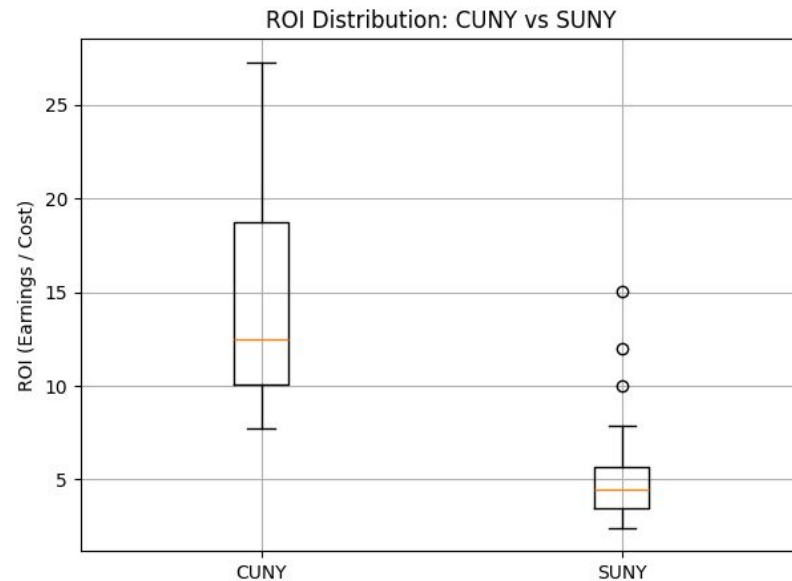
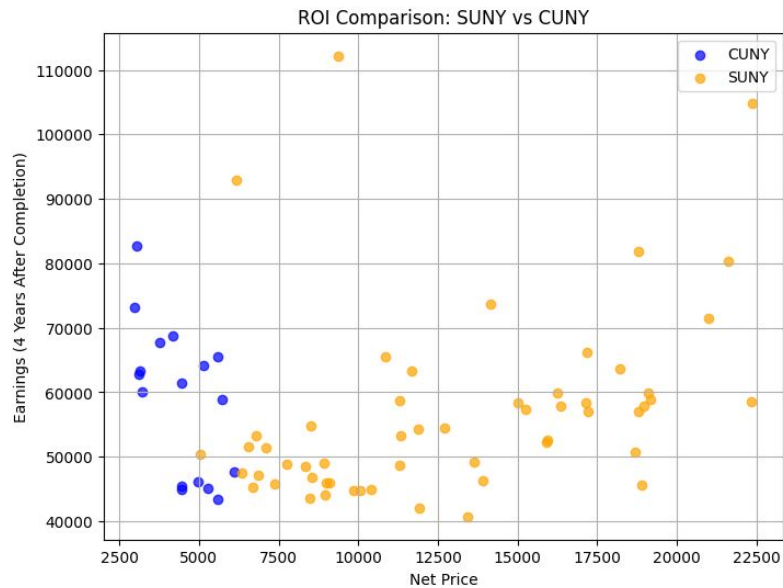
* Stony Brook

pell_% = 39

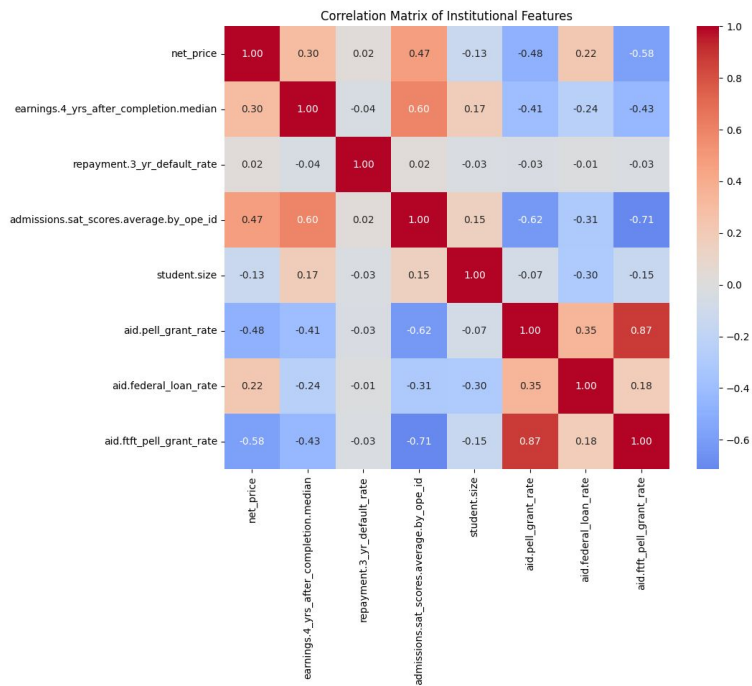
earnings = 81850.0

roi = 4.36

SUNY vs CUNY



Initial Insights



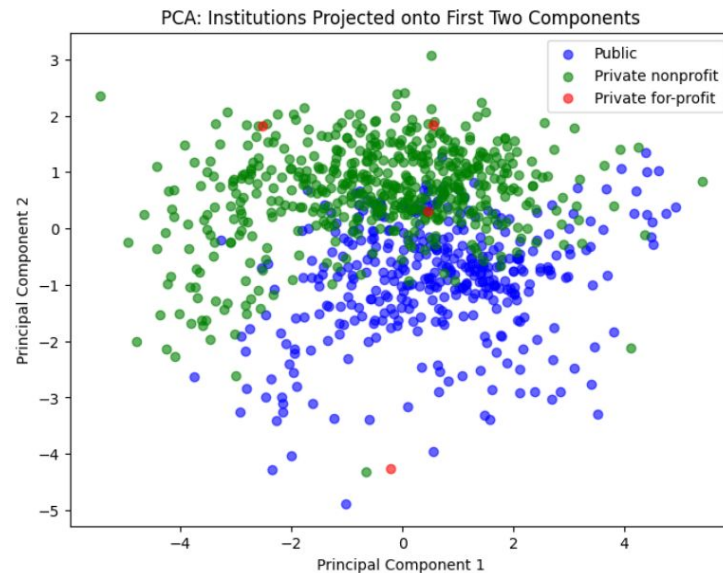
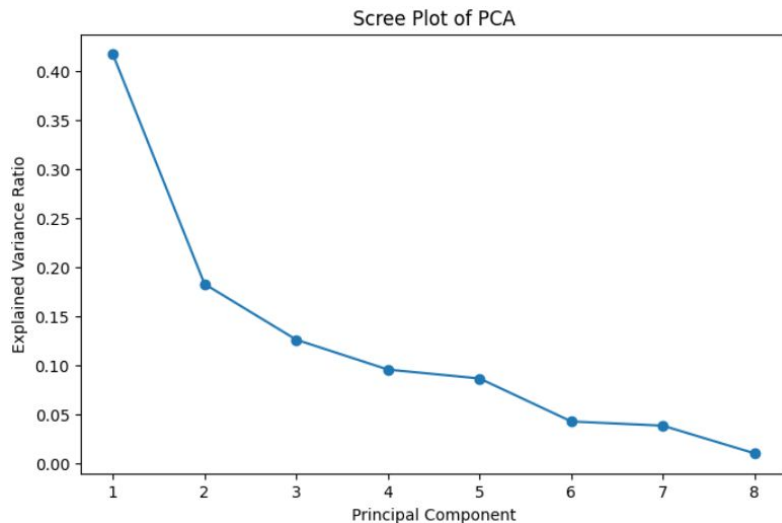
Do more expensive schools lead to higher salaries?

:No, net_price vs earnings $\approx$ 0.30 (weak)

Does institution size impact on certain features?

:No, student_size correlations are all weak

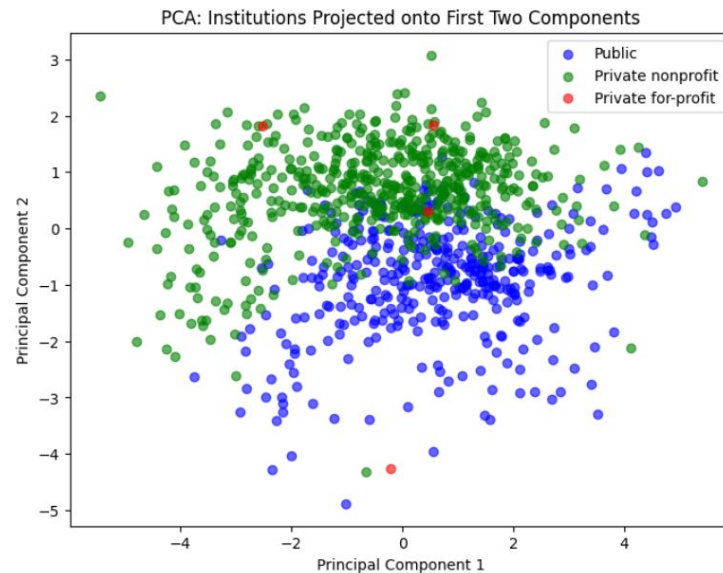
Lowering Dimensionality using PCA



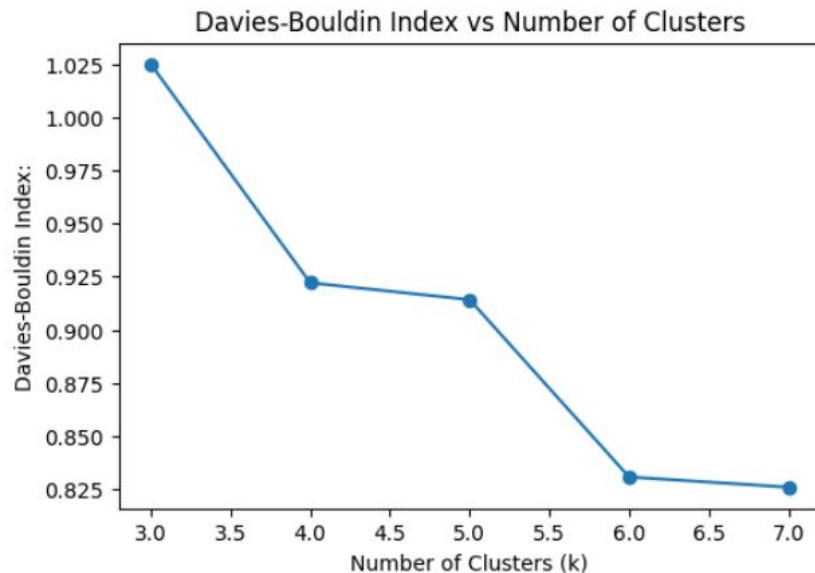
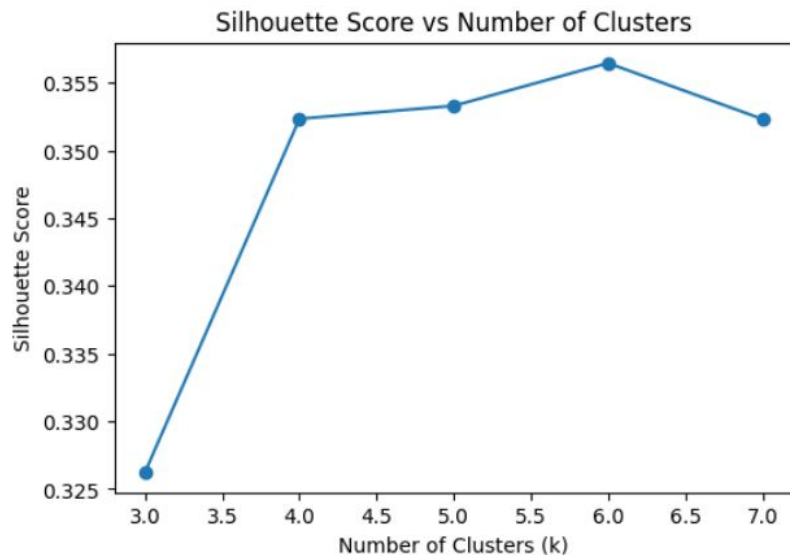
Lowering Dimensionality using PCA

	PC1	PC2
net_price	-0.339141	0.511859
earnings.4_yrs_after_completion.median	-0.367570	-0.121145
repayment.3_yr_default_rate	-0.014431	0.061766
admissions.sat_scores.average.by_ope_id	-0.473900	-0.031813
student.size	-0.110938	-0.564115
aid.pell_grant_rate	0.477603	-0.026210
aid.federal_loan_rate	0.191824	0.622540
aid.ffmpeg_grant_rate	0.497873	-0.109674

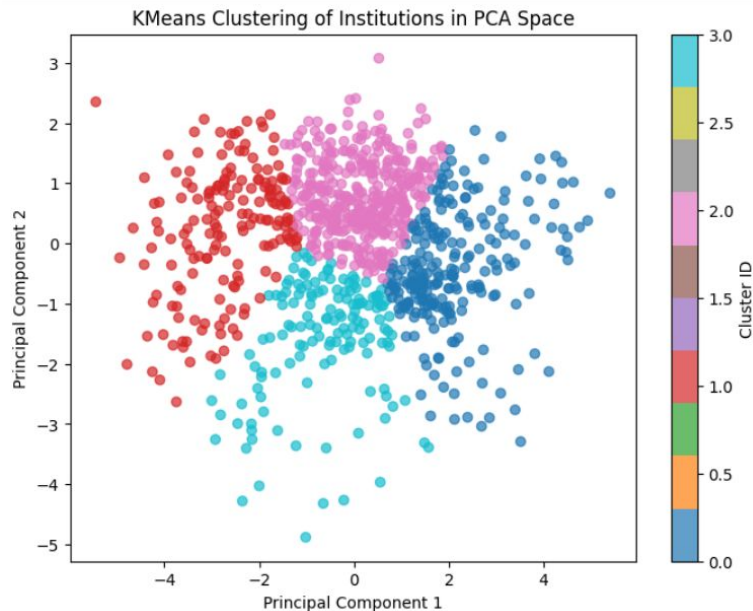
PCA Loadings



Using KMeans w/ PCA Mapping to Estimate Groups



Using KMeans w/ PCA Mapping to Estimate Groups



K Means with $k = 4$

- Data is grouped into 4 relatively equal quadrants with some overlap around the center
- Indicative that the problem is more nuanced (ex. just because a University is Public does not necessarily mean its earning potential is lower)

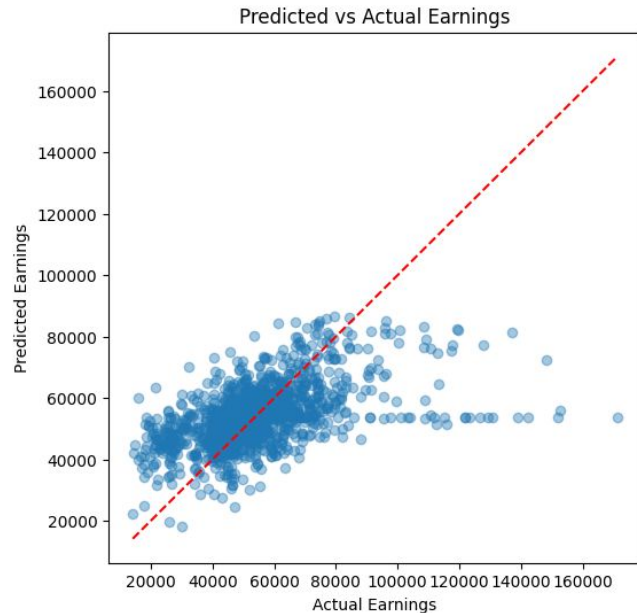
How Representative is the Estimation

	net_price	SAT	Earnings	Default Rate	Pell Grant R	Loan Rate	Student Size
cluster							
0	13741.337607	1044.80	55451.61	0.0	0.45	0.43	4624.41
1	32707.255814	1388.37	81701.67	0.0	0.17	0.33	5785.08
2	23642.844687	1176.52	61968.52	0.0	0.30	0.55	2152.85
3	15625.173611	1225.39	68884.41	0.0	0.26	0.30	16973.90

* Stony landed in the same cluster with Binghamton, UBuffalo, and UAlbany!

- Group 3 consisted of large public universities with more than 10,000 students
- All Ivy League colleges landed in the same cluster 😲
- Groups 0 and 3 tended to have the lowest net price
- Schools in group 1 significantly higher median earnings with an average median of 81,701.67
- Groups seem largely indicative

Earning Prediction Regression Model



Input = net price, pell grant and federal loan rate, and admissions sat score average

Output = Earnings 4 years after completion

Model Performance

- RMSE: ~\$16.4K
- R^2 : ~0.26

Factors beyond institutional characteristics provided in this dataset (major, location, and individual ability) play a significant role



Rethinking the Values of Higher Education

Expensive $\neq$ Better

ROI varies significantly across institutions

Hidden structure exists in higher education (revealed through clustering)

These findings are not universally applicable, as many other factors also influence outcomes

We would want to try to look into more resources(degrees_rewarded / retention_rates)

to find more meaningful correlations